

Database Systems Final Project Report

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Problem Description

Project Overview

This project implements the core data model of a small online store. The PostgreSQL 16 schema *shop* supports essential workflows: customer registration, product management, order creation, payments, and shipments. The design follows 3NF and relies on clear primary/foreign key relationships to enable accurate reporting and revenue analysis.

Users

The system serves customers, store administrators, accounting staff, and logistics personnel, all accessing the data through an application layer.

Entities

Six normalized entities form the model: Customer, Product, Order, Order_Item, Payment, and Shipment. Each table uses a surrogate primary key and stores only attributes relevant to its role.

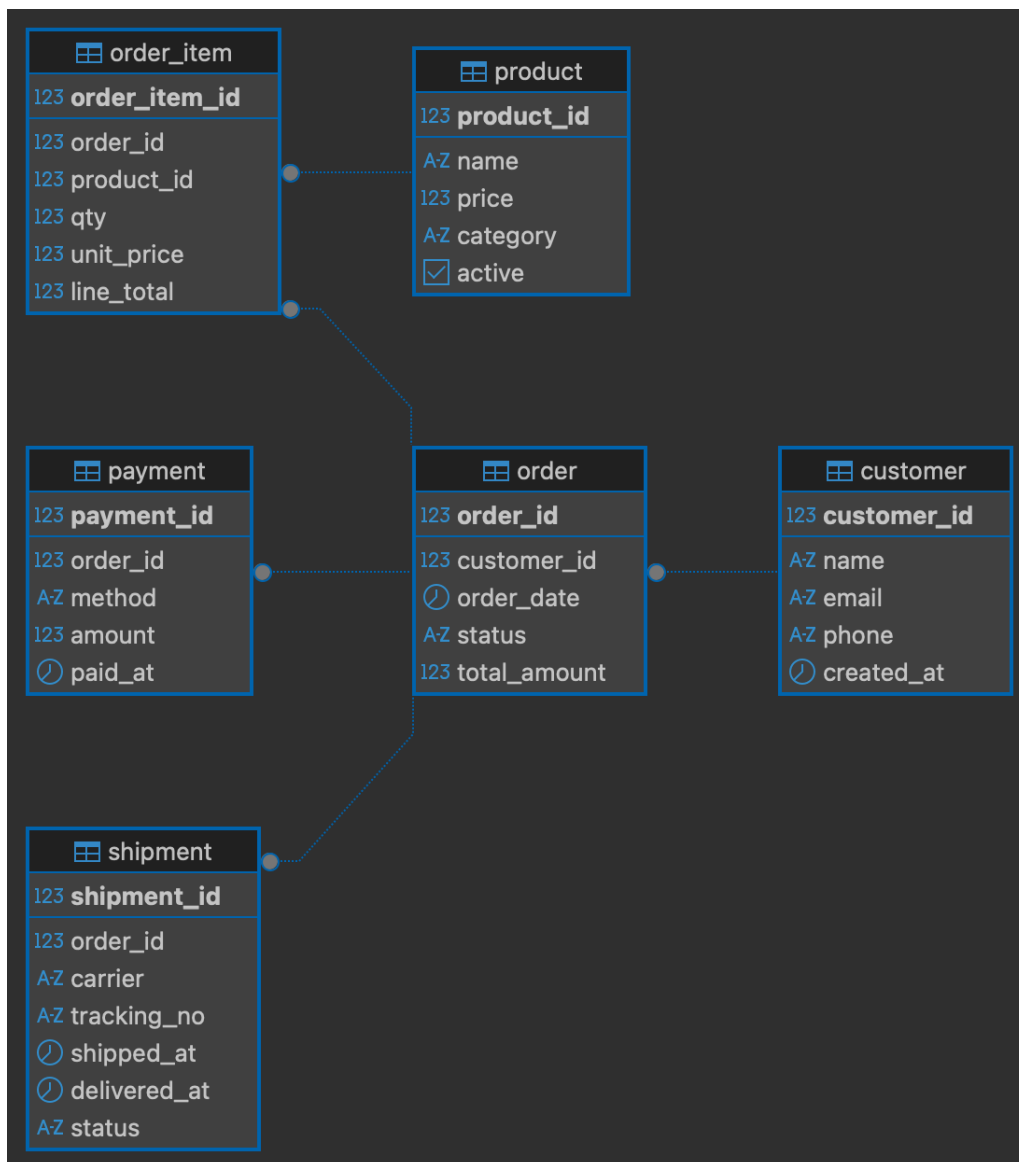
Relationships

The schema uses standard 1:N relationships: customers -> orders, orders -> order items, products -> order items. Orders may include multiple payments and shipments, all enforced through foreign-key constraints.

Business Rules

Core constraints ensure consistency: every order requires a customer and at least one item; line totals must equal quantity x unit price; prices and quantities must be non-negative; payments and shipments cannot exist without an order. Together these rules create a stable and realistic foundation for online-store operations.

ERD

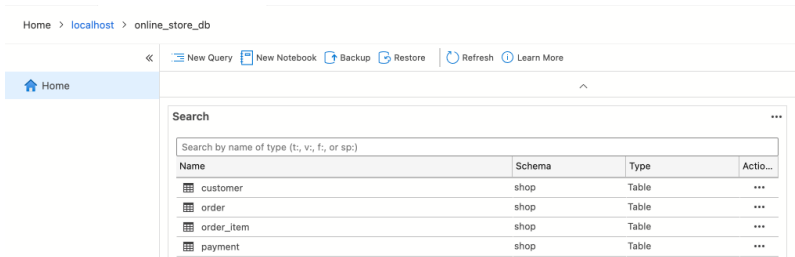


ERD Overview

The ERD shows six entities linked through primary and foreign keys, using Crow's-foot notation to represent all core 1:N relationships within the online-store schema. Customers can place many orders; each order contains items tied to specific products, and may include multiple payments or shipments that all depend on an existing order.

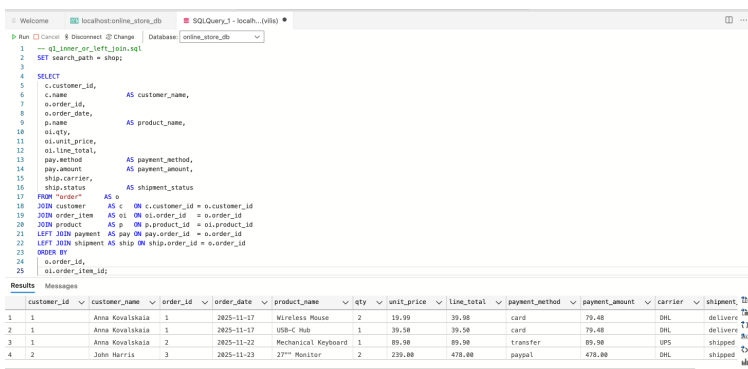
The schema conforms to 3NF, storing only atomic, non-redundant data; product details remain in *product*, while *order_item* keeps the unit price used at sale.

Screenshots & Comments



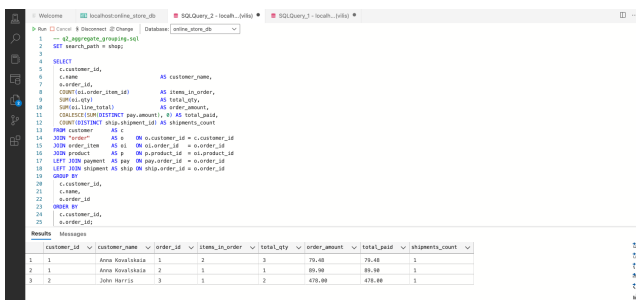
This screenshot confirms that the schema shop exists in the *online_store_db* database and all six tables are created and populated with sample data

Query 1 – Multi-table JOIN



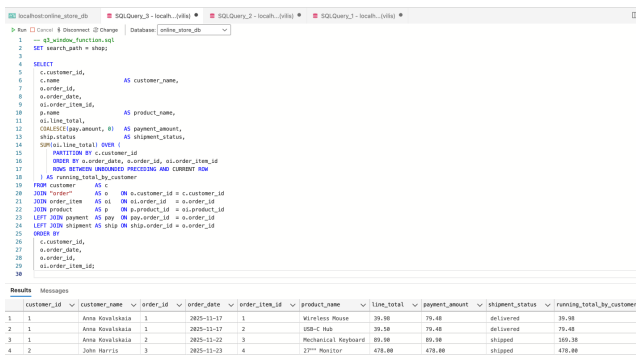
Screenshot shows an INNER JOIN combining order, customer, and payment data into a single view. Importance: verifies FK integrity and supports order-management and invoice screens.

Query 2 – Aggregation & GROUP BY



Screenshot demonstrates SUM() with GROUP BY to compute item counts and total amounts per order. Importance: essential for sales reporting and confirms correctness of line_total calculations.

Query 3 – Window Function (Running Total)



Screenshot illustrates a running revenue total using SUM() OVER(...). Importance: showcases advanced SQL analytics and progressive order-value calculation.